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Lecture Notes «Matemātiskā analīze I»

CALCULUS I

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LECTURE 7.

FUNCTION ANALYSIS I

Minimum and maximum values of function
Monotonicity, increasing and decreasing functions
7. Homework

Minimum and maximum values of function

Often in life, we are faced with the problem of finding the best way to do something. For example, a farmer wants to choose a mix of crops that is likely to produce the largest profit. A doctor wishes to select the smallest dosage of a drug that will cure a certain disease. Often such a problem can be formulated so that it involves maximizing or minimizing a function over a specified set. If so, the methods of calculus provide a powerful tool for solving the problem.

DEFINITION 7.1.

Let *S*, the domain of *f*, contain the point *c*. We say that:

1) *f(c)* is the maximum value of *f* on *S* if $f(c) \geq f(x)$ for all *x* in *S*

2) *f(c)* is the minimum value of *f* on *S* if $f(c) \leq f(x)$ for all *x* in *S*

3) *F(c)* is an extreme value of *f* on *S* if it is either the maximum value or minimum value

Minimum and maximum values of function

If f is <u>continuous</u> on a <u>closed interval</u> $[a, b]$, then f attains both a maximum value and a minimum value there.

How to find the points, where the function attains these extreme values?

Let f be defined on an interval I containing the point c. If $f(c)$ is an extreme value, then c must be a critical point; that is, either c is:

1) an end point of I;
2) a stationary point of f; that is, a point where $f'(c) = 0$; or
3) a singular point of f; that is, a point where $f'(c)$ does not exist.

Minimum and maximum values of function

FERMAT'S THEOREM

(necessary condition for the existance of the extreme value)

Let f be defined on an interval I. If a differentiable function attains its extreme value at the point c anc c is not an end point, then $f'(c) = 0$.

Should be taken into account that, the necessary condition does not guarantee that at a point where the derivative is 0 the extreme value will be attained.

Therefore, for finding the maximum and minimum value of a continuous function f on a closed interval I:

1) Find the critical points of f on I,
2) Evaluate f at each of these critical points. The largest of these values is the maximum value; the smales is the minimum value.

Minimum and maximum values of function

Example

Please, find extremal values of the function $f(x) = 2x^3 - 24x + 1$ on the interval [-3;1].

Find the derivative $f'(x) = 6x^2 - 24$, equal it to 0 and find two stationary points -2 and 2, but $2 \notin [-3; 1]$. Thus, the critical points are -3, -2 and 1.
Find values of the function at these points.
$f(-3) = 2 * (-27) + 72 + 1 = 19,$
$f(-2) = 2 * (-8) + 48 + 1 = 33,$
$f(1) = 2 - 24 + 1 = -21.$

We can conclude that the maximum value of the function $f(x) = 2x^3 - 24x + 1$ on the interval [-3;1] is $f(-2) = 33$, but the minimum value is $f(1) = -21$.
Answer:
$\max\limits_{[-3;1]}(2x^3 - 24x + 1) = f(-2) = 33,$
$\min\limits_{[-3;1]}(2x^3 - 24x + 1) = f(1) = -21.$

Exercise

Please, find extremal values of the function $f(x) = \frac{\sqrt{x^2-9}}{x^2-2}$ on the interval [3;5].

Monotonicity

DEFINITION 7.3.

Let *f* be defined on an interval *I* (open, closed, or neither). We say that:

1) *f* is increasing on I if, for every pair of numbers x_1 and x_2 in *I*

$$x_1 < x_2 \Rightarrow f(x_1) < f(x_2);$$

2) *f* is decreasing on I if, for every pair of numbers x_1 and x_2 in *I*

$$x_1 < x_2 \Rightarrow f(x_1) > f(x_2);$$

3) *f* is strictly monotonic on I if it is either increasing on I or decreasing on *I*.

Monotonicity

MEAN VALUE THEOREM FOR DERIVATIVES

If f is continuous on a closed interval $[a, b]$ and differentiable on its interior (a, b), then there is at least one number c in (a, b) where

$$\frac{f(b) - f(a)}{b - a} = f'(c).$$

FERMAT'S THEOREM and MEAN VALUE THEOREM FOR DERIVATIVES are called the basic theorems of the differential calculations in the basic course of calculus.

Monotonicity

THEOREM 7.4. (Monotonicity theorem)
Let *f* be continuous on an interval *I* and differentiable at every interior point.
1) If $f'(x) > 0$ for all x interior to *I*, then *f* is increasing on *I*.
2) If $f'(x) < 0$ for all x interior to *I*, then *f* is decreasing on *I*.

□ Let's suppose that $\forall x \in (a; b)\ f'(x) > 0$ and le'ts choose two arbitrary points $x_1, x_2 \in (a; b)$ such that $x_1 < x_2$.
In accordance with mean value theorem for derivatives
$\exists c \in (x_1, x_2)\ f'(c) = \frac{f(x_2)-f(x_1)}{x_2-x_1}$.
As for every $x \in (a; b)\ f'(x) > 0$, then also $f'(c) > 0$. What means that $\frac{f(x_2)-f(x_1)}{x_2-x_1} > 0$.
As $x_1 < x_2$, that is, $x_2 - x_1 > 0$, then we have to conclude that $f(x_1) < f(x_2)$, which means that function *f(x)* is increasing on interval *(a; b)*.
Similarly can prove the 2).■

Monotonicity

Example

Please, detect the increasing and decreasing intervals of the function $f(x) = \frac{4x+5}{1-x^2}$.

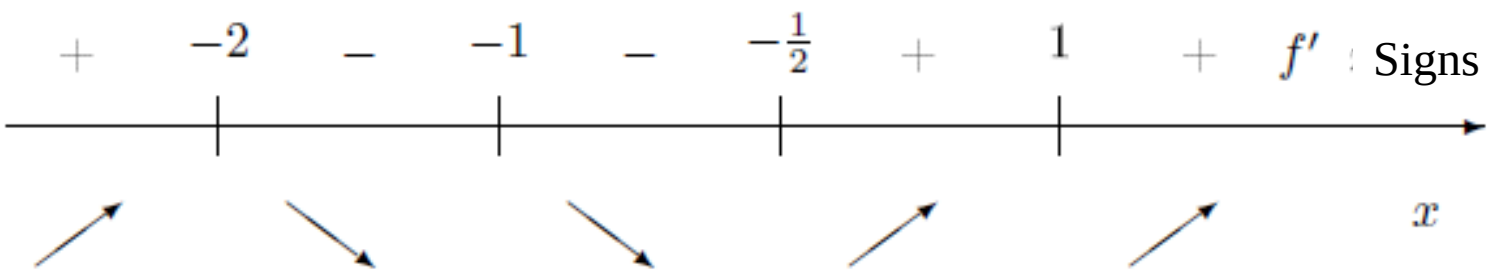
Let's find the derivativ $f'(x) = \frac{4(1-x^2)+2x(4x+5)}{(1-x^2)^2} = \frac{2(2x^2+5x+2)}{(1-x^2)^2}$

Let's find the critical points:

1) Stationary points (s where f'(c) = 0) – by finding roots of the equation $2x^2 + 5x + 2 = 0$.

$x_1 = -2 \; and \; x_2 = -\frac{1}{2}$;

2) Singular points (where *f'(c)* does not exist) – $x_3 = -1 \; and \; x_4 = 1$.



Answer: Increasing intervals $(-\infty; 2), \left(-\frac{1}{2}; 1\right) \; and \; (1; +\infty)$ and decreasing intervals $(-2; 1) and \; (-1; -\frac{1}{2})$

Monotonicity

NOTE

Cannot merge the increasing (or decreasing) intervals in one set, because in such merged set the function may not be increasing (or decreasing)!

Exercise

Please, detect the increasing and decreasing intervals of the functions $f(x) = \frac{x^2}{1-x}$ and $f(x) = \ln(x^2 + x - 6)$!

HOMEWORK 7

13. Please, for the given function $f(x) = \sqrt[3]{x^2} + \frac{2}{3}x$ find

a) the critical points,

b) the minimal amd maximal value at the interval $\left[-8; \frac{1}{8}\right]$,

c) the increasing and decreasing intervals of the function.

Answers:

a) -8, -1, 0 and $\frac{1}{8}$

b) $f(-8) = -\frac{4}{3}$ - minimalvalue; $\mathrm{f}(-1) = \mathrm{f}\left(\frac{1}{8}\right) = \frac{1}{3}$ - maximal value

c) increasing interval]-8; -1[and]0; $\frac{1}{8}$[, decreasing terval]-1;0[

14. The area of the cardboard page is 8 cm x 5 cm. By cutting out identical squares from the four corners and turning up the sides should be made a box with the maximum volume. Please find the length of the square side that shoud be cut out from the corners.

Answer: 1 cm

Reading: D. Varberg, E. J. Purcell, S. E. Rigdon *Calculus*. 9th Edition, Prentice-Hall, Inc.m, 2014, p. 151– 156, 167 – 168, 185 – 188.

Chapter 3. Sections 3.1., 3.2., 3.4., 3.6.